

The empirical recurrence for OEIS A254087, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A254087 records “Empirical recurrence of order 96” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 188$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 100$. Its last coefficient is $c_{96} = 2339171884152397435280588361695232 \neq 0$, so no further tail satisfies anything shorter; any order-96 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A254087 is “Number of $(n+2) \times (3+2)$ 0..1 arrays with every 3×3 subblock diagonal maximum minus antidiagonal maximum nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

22728, 437412, 9231485, 194642351, 4074710828, 85580284203, 1797518333568, 37753559435660, ...

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:27:21, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 188$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 188$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 96: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 96.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 376$ exact terms from $n = 4$ on, it returns order 96 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	25	c_{25}	−2873981855456869376	c_{49}	−194604061417652666630269829120
c_2	74	c_{26}	−19028584596400902144	c_{50}	−130793466105753551804041265152
c_3	−3922	c_{27}	−62015750932079628288	c_{51}	−879838857739473878395771682816
c_4	8499	c_{28}	609012396045575913472	c_{52}	−707809685318819623430468403200
c_5	100677	c_{29}	146083501910781640704	c_{53}	2635658927988553283137563525120
c_6	−438272	c_{30}	4395038710391680139264	c_{54}	1512234610686919608614216794112
c_7	4596776	c_{31}	337857196105811951616	c_{55}	15436163413082959858506303275008
c_8	−31381116	c_{32}	−30498487879440811687936	c_{56}	9190489925376020277003454251008
c_9	338829100	c_{33}	−37908451561667436478464	c_{57}	−2988769525204813177519849917644
c_{10}	290830160	c_{34}	−398823695144776492908544	c_{58}	−1575321675372609489568517849088
c_{11}	−37313273328	c_{35}	61188596834557006708736	c_{59}	−19626068723548311785830122677862
c_{12}	107202880752	c_{36}	487255220291998424498176	c_{60}	−8982098164137316935520481181696
c_{13}	278976895184	c_{37}	5313992572260430377910272	c_{61}	308196389311361626052829393190912
c_{14}	−1294945138176	c_{38}	18863079053700383167741952	c_{62}	12601429467938300458702508916736
c_{15}	28891041420672	c_{39}	1021744632157233123164160	c_{63}	175207667988514310565548239985049
c_{16}	−90861769333568	c_{40}	26206910822454959097774080	c_{64}	64838282160546351146921962032332
c_{17}	−410241650033088	c_{41}	−292273032179925956933189632	c_{65}	−18879100264685249981105147928903
c_{18}	945674896933120	c_{42}	−531755939556448952263901184	c_{66}	−79315682591042164578677671591936
c_{19}	−4432887649873664	c_{43}	−624697681159216410149257216	c_{67}	−123753450403189054815039437427179
c_{20}	27509595796634880	c_{44}	−1628753461578137772754993152	c_{68}	−37124751420163056930157446205276
c_{21}	28703553721845504	c_{45}	9351970182432968987818065920	c_{69}	760782516767092902410480588842598
c_{22}	−51880597362352128	c_{46}	9784043129989582239572164608	c_{70}	457756125531508455012005337628672
c_{23}	1051883582860617728	c_{47}	34149268177437205445071601664	c_{71}	607664368421088043386822669358858
c_{24}	−5894720728960627712	c_{48}	41910780417206576715204132864	c_{72}	165880526118726239790146192052060

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 100$, and it is the recurrence OEIS A254087 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{96} - c_1x^{96-1} - \dots - c_{96}$. Its constant term is $-c_{96} = -2339171884152397435280588361695232$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 96 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 96, so $\deg q = 96$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 96, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 96, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 2339171884152397435280588361695232.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-96 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A254087.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.